

WPP Enterprise Agent Framework — Solution Architecture

1. Principles

The determinism ↔ agentic spectrum

WPP's processes span a spectrum. Some phases are rule-driven and must be deterministic (three-way PO match, jurisdiction routing, payment file generation, IT provisioning tickets). Others require reasoning over unstructured input and judgement (CV triage, voice screening, compliance narrative review). A 12-week hiring process is not one mode or the other — it is a **sequence of deterministic scaffolding gates with agentic reasoning inside specific steps**.

Our architecture expresses this spectrum explicitly through three execution layers. Each has a different substrate, a different failure model, and a different cost profile. Mixing them is the point — not a compromise.

Three-layer execution model

Layer	Substrate	Role	Determinism
1. Durable execution envelope	Azure Durable Functions (GA)	One orchestration instance per workflow. Owns phase boundaries, HITL waits at zero compute (days or weeks), timer escalation, checkpoint/replay, geo-replicated state.	Fully deterministic — code-defined, event-sourced replay.
2. Workflow graph	Microsoft Agent Framework (MAF) workflows with the durable task extension for Azure Durable Functions	Defines each phase (or the whole process) as a graph of typed executors with fan-out/fan-in, conditional routing, and HITL hooks. Validator executors sit between agent executors.	Deterministic by default, agentic by exception — most nodes are plain functions.
3. Agent executor contents	GHCP SDK sessions running on Foundry Hosted Agents	Invoked from MAF agent executor nodes only. Load skills and MCP tools , reason, call tools through hooks , emit OTEL , exit.	Probabilistic LLM reasoning sandboxed inside a single executor, wrapped by deterministic pre/post hooks.

A WPP workflow is therefore a **Durable Functions orchestration** that coordinates **MAF workflow graphs** whose agent nodes invoke **GHCP SDK sessions**. Each layer earns its place: DF for long-running durability that MAF does not itself provide at zero compute; MAF for the deterministic graph shape of a phase with HITL and typed data flow; GHCP SDK for the LLM runtime with hooks, MCP, skills, and OTEL.

Deterministic by default, agentic by exception

Most executors in a MAF workflow are plain Python/C# functions — they fetch data, validate, route, emit events. Agent executors are used only where LLM reasoning is genuinely required. Between agent executors, **validator**

executors assert structural and policy constraints before output propagates — this is the "judge catches executioner" pattern made explicit in the graph, not delegated to the model.

The consequence for governance: the majority of the workflow graph is literally code. The LLM is contained inside specific executors, wrapped in GHCP SDK hooks (pre/post tool-use), with Foundry Guardrails intercepting at four points, with external validators between nodes, and with APIM AI Gateway governing every model and tool call. Probabilism is bounded.

Skill crystallisation — the migration path

Proven patterns move left along the spectrum. A phase that starts as an agent executor producing structured output, validated by a downstream executor, and found stable over N completed workflows can be **crystallised** — promoted from LLM-generated to deterministic code, versioned as a skill in [Azure API Center](#), and swapped into the MAF graph as a plain function executor. The agent executor remains available as a fallback for exceptions. This is how the system gets cheaper, faster, and more predictable as it matures — without re-architecting anything.

Why GHCP SDK for agent executors

The agentic loop pattern is well understood. The hard part is the runtime around it: session management, skill resolution, MCP client lifecycle, hook interception, OTEL instrumentation, model failover, structured outputs, sub-agent delegation, prompt caching. Building this from scratch is a multi-quarter engineering effort, and what you end up with is what GitHub already runs in production behind GitHub Copilot — serving millions of developers daily.

[GHCP SDK](#) is that runtime, extracted and open-sourced (MIT, Python/TypeScript/Go/.NET). Inside a MAF agent executor, adopting it means:

- **Scaled, not DIY** — the loop, hooks, skills, and MCP client implementations are battle-tested at GitHub Copilot scale, not bespoke code we maintain
- **Composable governance** — [hooks](#) intercept tool calls deterministically; non-revocable actions route to humans without LLM intervention in the send path
- **Open standards native** — [MCP](#) (tools), A2A (agent-to-agent), [OTEL](#) (observability) are first-class, not bolted on
- **Portable** — same SDK code runs on Foundry Hosted Agents, Container Apps, AKS, or local dev. No lock-in to a specific hosting fabric
- **Specialisation via skills, not via separate agents** — one domain-scoped Hosted Agent (with one Entra Agent ID, one tool allow-list, one audit identity) runs many phase-specific executors by loading different [SKILL.md](#) files. Avoids the "300 agents to manage" anti-pattern

Everything else in this document (Foundry, APIM, MAF workflows, DF, Fleet Managers, Control Plane, IQ products) is the **enterprise envelope** around these three layers.

2. Approach

Azure Durable Functions provides the long-running durable envelope around each workflow. Inside each phase, a **Microsoft Agent Framework (MAF) workflow** graph — wired to DF via the MAF [durable task extension](#) — defines the deterministic execution path, with most nodes as plain functions and agent nodes invoking **GHCP SDK** sessions on Azure AI Foundry Hosted Agents. The entire stack is governed by Fleet Manager agents, grounded by **Foundry IQ / Fabric IQ / Work IQ**, centrally managed through **Azure API Center + APIM AI Gateway**, and surfaced through Foundry Control Plane.

Refs: [GHCP SDK](#) (MIT, Python/TypeScript/Go/.NET) · [Microsoft Agent Framework](#) (v1.0, Python/.NET) · [MAF Durable Functions integration](#) · [Durable Agent Orchestration tutorial](#)

Skills: [SKILL.md files](#) define capabilities declaratively. New capability = new skill file. Skills are the specialisation mechanism — instead of separate agents for screening, sourcing, compliance, etc., a single domain-scoped Hosted Agent loads different skills per MAF agent executor. Each skill defines its own role, allowed tools, model assignment, and governance rules. Skills are also crystallisation artefacts — proven agentic patterns graduate from LLM-generated to deterministic code, versioned as skills, and can be swapped into the MAF graph as plain-function executors (agent executor kept as exception fallback). Skills are registered in Azure API Center with lifecycle management (Design → Preview → Production → Deprecated) and governed via APIM AI Gateway.

Ref: [Azure API Center lifecycle](#)

MCP tools: enterprise system integrations (Workday, Greenhouse, LinkedIn, ServiceNow, Graph, Dataverse) exposed as [MCP servers](#), governed through APIM AI Gateway with auth, rate limiting, and content safety policies. APIM provides a [REST-to-MCP gateway](#) that auto-generates MCP tool definitions from OpenAPI specs.

Hooks: GHCP SDK session hooks ([onPreToolUse](#) , [onPostToolUse](#)) provide operational governance inside each agent executor's session. Non-revocable actions (send email, submit background check, extend offer, execute payment) are intercepted by hooks, blocked from immediate execution, and routed to the human's PA for approval. After approval, execution is deterministic — no LLM in the send path. Hooks also handle audit logging (every tool call logged with workflow context) and tool allow-listing per skill set. Hooks complement MAF validator executors: hooks operate inside the session; validators operate on its typed output.

Human interaction: humans interact with their personal agent (PA) through M365 Copilot (Teams, Outlook) via the [M365 Agents SDK](#), and through the custom Control Plane UI for fleet management. There is always an agentic layer between humans and the system.

3. Architecture

Layers

Layer	Component	Role
Fleet Manager	Always-on GHCP SDK Hosted Agent	Consumes telemetry from all workflows. Reasons about fleet health, SLA risk, anomalies. Composes the exception queue for the Control Plane. Monitors compliance enforcement events.
Durable envelope	Azure Durable Functions (GA)	One orchestration instance per workflow. HITL waits at zero compute for days or weeks, timer escalation, checkpoint/replay, geo-replicated state. Coordinates phase boundaries and escalations.
Workflow graph	MAF workflow via durable task extension	Deterministic graph of typed executors for each phase. Plain function executors for data mapping, validation, routing. Agent executors where reasoning is required. Validator executors between agent executors (judge/executor separation). Pregel BSP execution, fan-out/fan-in, conditional routing, HITL hooks.
Agent executor contents	Ephemeral GHCP SDK sessions	Invoked only from agent executor nodes. Load skills + MCP tools. Reason about the task. Write state. Emit OTEL telemetry . Exit.

Identity & Access

The Entra Agent ID lives on the **Hosted Agent container**, not on individual sessions. The GHCP SDK session invoked from a MAF agent executor is a unit of work inside a persistent container — it uses the container's identity when

calling tools.

Hosted Agent topology (design decision for POC):

Hosted Agent	Entra Agent ID	Tool Access	Hosts Agent Executors For
Hiring Agent	hiring-agent@wpp	Greenhouse, LinkedIn, Workday (hiring), Graph, ACS	All hiring workflow agent executors (screening, sourcing, compliance, offer, onboarding)
Finance Agent	finance-agent@wpp	Workday (finance), D365 F&O, Maconomy	All P2P workflow agent executors (intake, validation, routing, payment, reconciliation)
Fleet Manager	fleet-manager@wpp	Read-only: Foundry Tracing, Event Grid, workflow state store. No tool invocation on downstream systems.	Fleet monitoring, exception composition, compliance oversight

Each Hosted Agent is domain-scoped — it has only the tool access its domain requires. All GHCP SDK sessions invoked from MAF agent executors inside it inherit that scope.

OBO vs app-only: when a MAF workflow is triggered by a human action (Finance BP approves), the GHCP SDK session inside an agent executor can act [on-behalf-of](#) that human — audit trail attributes the decision to them, downstream access uses their delegated permissions. For autonomous phases (screening 200 CVs), the session uses the Hosted Agent's app-only identity.

Skill and policy governance: skill promotion (Design → Production) goes through Azure API Center lifecycle gates. API Center integrates with GitHub Actions workflows for syncing SKILL.md files from repositories. Autonomy threshold changes are audit-logged with operator identity. Governance policy changes require PR review in Git, deployed via [APIOps CI/CD](#).

Human Interaction Model

Every human has a **personal agent (PA)** — a GHCP SDK agent surfaced in M365 Copilot via M365 Agents SDK. The PA knows the human's role, permissions, and context. It surfaces information, recommends actions, drafts outputs, and triggers workflows on behalf of the human. Humans make all decisions. The PA prepares and executes.

Surface	How	Who
M365 Copilot (Teams)	Each user's PA agent. Understands role context, surfaces relevant information proactively, triggers workflows OBO the user when instructed.	All WPP users
Email (Adaptive Cards)	PA composes Adaptive Card with context and recommendation. Human reviews and decides. Response routes back through PA.	Finance BP, approvers
Control Plane UI	A richer surface the PA uses when the situation requires more than a Teams message — fleet view, bulk approvals, drill-downs. The HR BP's PA IS the Fleet Manager.	HR BP, Fleet operators
ServiceNow	PA writes provisioning tasks via ServiceNow MCP.	IT Ops
Web portal	Custom web surface for external participants.	Candidates

Voice	GPT-Realtime as speech-to-speech front end. Tool calls to GHCP SDK backend for reasoning.	Candidates (screening)
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Flow (Hiring Workflow)

- 1. **Hiring Manager's PA** surfaces a headcount gap based on Workday/Dataverse data: "There's an open Senior Data Engineer role. I've prepared a draft requisition — want me to kick off the hiring workflow?"
- 2. **Hiring Manager** reviews and approves. PA triggers the Durable Functions orchestration, acting OBO the manager.
- 3. **Durable Functions** starts the Budget & Job Design phase by invoking the corresponding MAF workflow as a durable activity. The MAF graph runs: a deterministic `fetch_headcount_context` executor pulls Workday/Dataverse data, an `agent_job_design` executor (GHCP SDK session with job-design skill) drafts the JD, a `validate_jd_schema` executor asserts structure, a `compute_budget_envelope` function finalises numbers. State is emitted back to DF.
- 4. **Durable Functions:** budget needs Finance BP approval → `wait_for_external_event` (zero compute, preserved across MAF workflow checkpoints via the durable task extension)
- 5. **Finance BP's PA** receives the approval request, presents it in Teams or via Adaptive Card with context and recommendation. Finance BP reviews and approves.
- 6. **PA** processes the decision, writes to state → `raise_event` → Durable Functions resumes and invokes the next-phase MAF workflow
- 7. **HR BP's PA** (Fleet Manager) monitors all workflows, proactively surfaces: "2 of your 20 workflows need attention. Workflow-456 has a compliance flag — here's what happened and my recommendation."
- 8. **HR BP** acts on the recommendation via Control Plane UI or directly in Teams. The other 18 workflows run autonomously.

4. API Centre & AI Gateway

[Azure API Center](#) and [APIM AI Gateway](#) together form a **single, cohesive control plane for everything addressable by an agent** — models, MCP tools, A2A agents, skills, and APIs. One pane, one policy engine, one audit trail.

What flows through this layer

Asset class	API Center role	APIM AI Gateway role
LLM endpoints (Foundry models, third-party)	Model registry, version metadata, jurisdiction tags	Load balancing & failover , spillover to alt regions, token rate limits , semantic caching , cost emission , per-team/per-workflow budget control
MCP tools	Tool catalogue with metadata, lifecycle, allowed-skill matrix	Auth injection, rate limiting, content safety , discovery , REST→MCP auto-generation
A2A agents	Agent card registration, capability discovery	JSON-RPC mediation , policy enforcement , SSE/gRPC transport (preview)
Skills	SKILL.md lifecycle (Design → Preview → Production → Deprecated), GitHub Actions sync, allowed-tools governance	Per-skill tool allow-list enforcement at runtime

APIs (REST, OpenAPI)	OpenAPI registry, cross-cloud discovery (Azure / GCP / AWS / on-prem)	Standard APIM governance: auth, throttling, transformation
Foundry IQ knowledge bases	Registered as discoverable retrieval endpoints	Uniform auth + audit for grounding calls

Why this matters for WPP

- **Cost control at scale:** 500 concurrent workflows × 30 markets means token spend is operational risk. APIM's per-model/per-team/per-workflow token metrics + budget enforcement is the only sane way to cap this. (Refs 26.1, 26.4)
- **Cross-cloud discovery:** API Center registers MCP servers, skills, and APIs hosted **anywhere** — Azure-hosted Foundry models, GCP-hosted Vertex agents, AWS-hosted internal APIs, on-prem SAP. Single registry, single discovery endpoint for agents. (Refs 8.13, 22.4, 33.3)
- **Jurisdiction routing:** APIM routes a German workflow to an EU-only model endpoint without the agent needing to know the model topology. Policy lives in the gateway, not the agent code. (Ref 32.1)
- **DR & spillover:** model-level load balancing across regions, automatic failover when a region degrades, no agent restarts needed.
- **One auditable record** of every model call, tool call, A2A handoff, and skill invocation — feeds Log Analytics for the 7-12 year retention requirement. (Ref 10.5, 32.3)
- **Policy-as-code:** APIM policies version-controlled via [APIOps CI/CD](#). Skill and tool registrations in API Center synced from Git. Governance changes go through PR review. (Ref 21.4)

This addresses **Refs 8.13, 8.16, 22.1-22.6, 26.1-26.4, 32.1-32.3** in one architectural primitive.

5. Intelligence Layer

Agents need three kinds of grounding: **enterprise documents** (policy, law, SOPs), **business semantics** (what entities exist, how they relate, what rules apply), and **work context** (who collaborates with whom, what's currently in flight). Microsoft now has a purpose-built product for each, and all three are MCP-addressable — meaning they sit behind APIM AI Gateway alongside any other tool, governed identically.

Product	Role	Status	Skills/Agents that consume it
Foundry IQ	Unified agentic retrieval over enterprise corpora. Self-reflective query planner with configurable "retrieval reasoning effort". Federates SharePoint, Fabric, OneLake, Blob, AI Search, web, and MCP behind one endpoint. Permission-aware. Built on Azure AI Search.	Public preview	Compliance Agent (right-to-work, EU AI Act, GDPR), Job Design (comp benchmarking, role profiles), Skill Amplification (Fleet Manager surfacing precedents and policy to operator), Onboarding (Day-1 plan templates)
Fabric IQ	Semantic intelligence layer over WPP's Fabric / OneLake estate: business ontology (entities + relationships + rules), semantic model (BI definitions extended to AI), graph engine for multi-hop reasoning, data agents and operations agents .	Public preview	Budget Agent (headcount, cost-centre, agency hierarchy), ROI reporting , Process Evolution (detect inefficiencies after N workflows), cross-entity reasoning (matrix navigation: role in Media, budget from WPP Corp)

Work IQ (MCP)	M365 work graph + memory layer behind Copilot. Three layers: data (M365, D365, Power Platform), context/memory (collaboration patterns, expertise, work velocity), skills & tools. Exposed to custom agents via MCP servers.	Preview	Personal Agents (every PA's grounding for "who is this person, what do they care about"), Interview Coordinator (timezone, availability, calendar patterns), Org Topology / Escalation (who has authority, who's reachable, who's the SME)
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How agents use them

- A loop loads a skill (e.g. `compliance.right-to-work-DE`)
- The skill declares an MCP tool dependency on `foundry-iq:de-employment-law` (a knowledge base registered in Foundry IQ)
- The loop's tool call goes through APIM AI Gateway, which authenticates via the Hosted Agent's Entra Agent ID, applies content safety, logs to App Insights, and forwards to Foundry IQ
- Foundry IQ plans a multi-source retrieval (BetrVG corpus + GDPR consent guidance + WPP DE handbook), synthesises with citations, returns
- The loop reasons over the result and acts

The **substrate** for Foundry IQ is Azure AI Search, but agents address Foundry IQ, not raw AI Search — the agentic retrieval planner is the value-add.

Mapping to WPP requirements

Ref	Requirement	Covered by
6.3	Learning from past interactions and human feedback	Work IQ memory + Fabric IQ ontology updates
16.1	Tiered memory	Work IQ (working/episodic) + Foundry IQ (semantic/factual) + Fabric IQ (procedural/business rules)
16.4	Grounding in knowledge bases, document stores, databases, APIs	Foundry IQ (one endpoint, all sources)
23.1	Process mining from email, docs, system interactions, meetings	Work IQ
25.1-25.3	Org topology, escalation, cross-entity navigation	Work IQ + Fabric IQ graph engine
28.1, 28.3	Document comprehension + quantitative reasoning	Foundry IQ + Fabric IQ

6. Fleet Manager Agents

Domain-scoped GHCP SDK Hosted Agents that govern other agents. WPP cannot manage fleet-scale agentic operations relying on human eyes alone — agentic governance is required.

Multiple instances by domain: hiring fleet manager, finance fleet manager, compliance fleet manager. Each scoped to its domain's telemetry and workflows. If one degrades, its domain falls back to unfiltered view while others continue.

Inputs: telemetry events from DF orchestrations, MAF workflow executors, and GHCP SDK sessions via Event Grid. Enforcement events from Foundry Guardrails, MAF validators, and APIM.

Outputs: fleet health assessment, prioritised exception queue, contextual recommendations, compliance alerts — pushed to Control Plane UI via SignalR.

What the operator sees: Fleet Manager's assessment. Exception-only view — only flagged workflows are visible. For each: what happened, why it stopped, recommended action, options. Bulk approval queue for batched low-risk decisions. Autonomy dials for per-workflow thresholds.

Delivers WPP's Refs 31.1-31.5: exception-only surfacing, situational awareness, skill amplification, AI-driven prioritisation.

7. Workflow Durability & Execution

Three cooperating substrates, connected by Microsoft's [Durable Agent Orchestration](#) pattern (Feb 2026): Durable Functions + MAF + SignalR. This is the Microsoft-productised approach for long-running agent workflows with HITL waits.

Azure Durable Functions — the outer envelope: event routing across phases, HITL waits (zero compute, days/weeks), timer escalation, parallel coordination, checkpoint/replay, geo-replicated state. Durable Functions does **not** execute reasoning, tool calls, or skills — it coordinates.

MAF workflow — the per-phase graph: each phase is a MAF workflow graph of typed executors. The [MAF durable task extension for Azure Functions](#) lets a DF orchestrator invoke a MAF workflow as a durable activity. MAF's own checkpointing is preserved across DF replay — conversation context and workflow state persist automatically. Most executors are plain Python/C# functions (deterministic). Agent executors delegate to GHCP SDK sessions. Validator executors enforce structural and policy constraints between nodes.

GHCP SDK sessions — inside agent executors: ephemeral sessions that load skills and MCP tools, reason, call tools through hooks, emit OTEL, and return a typed result to the MAF executor.

Workflow state store ([Cosmos DB](#) / Dataverse): phase state, context, candidate data, approval records, append-only action ledger (revocable/non-revocable tracking), OTEL span summaries. Cosmos DB provides [multi-region writes with automatic failover](#) and [continuous backup with point-in-time restore](#).

HITL pattern: a MAF workflow executor detects human input is needed → composes contextual message/Adaptive Card → routes it to the appropriate human via their preferred surface → signals DF to suspend. DF issues `wait_for_external_event` at zero compute. Human responds → response triggers `raise_event` → DF resumes, invokes the resumption MAF workflow to process the decision → next phase begins. Bulk approval raises events on multiple DF instances simultaneously. MAF workflows also support their own native pause/resume for shorter-lived HITL within a phase ([MAF human-in-the-loop](#)).

Execution model: MAF workflows use a Pregel-based Bulk Synchronous Parallel (BSP) execution model — supersteps collect pending messages, route to target executors per edge definitions, run all target executors concurrently, wait for completion before advancing. This makes parallel phases and fan-out/fan-in deterministic and inspectable. Combined with MAF's stable [orchestration patterns](#) (sequential, concurrent, handoff, group chat, Magentic-One), this covers the multi-agent topologies WPP requires without bespoke coordination code.

8. Observability

Layer	Source	Purpose
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Foundry Tracing (OTEL)	Every GHCP SDK session (inside MAF agent executors) emits via GHCP SDK OTEL TracerProvider. MAF workflow execution spans are emitted natively.	Raw telemetry: executor lifecycle, model calls, tool calls, tokens, latency, cost. Drill-down for situational awareness. Foundry Agent Monitoring Dashboard out-of-box.
APIM AI Gateway metrics	Token usage , latency, errors per model/tool/agent	Central cost tracking and rate limit monitoring across all AI assets.
Fleet Manager assessment	Event-driven reasoning over telemetry	Intelligent layer: fleet health, anomaly detection, SLA risk, exception prioritisation. Default Control Plane view.

OTEL spans carry workflow ID, phase, jurisdiction, model, token count. Cost attribution per workflow, phase, model, and consumer visible in Control Plane and Foundry dashboards.

9. Compliance & Governance — Four Enforcement Layers

Compliance is enforced externally and structurally. The LLM cannot bypass these.

Layer	Technology	Status	Enforcement
MAF workflow validators	Validator executors in the MAF workflow graph between agent executors	GA (MAF v1.0)	Structural and policy enforcement inside the graph. A validator executor downstream of an agent executor asserts schema, checks against policy tables, runs deterministic rule engines, and routes to a rejection branch if violated. Judge/executor context separation is expressed as two nodes and an edge — the judge does not share the executioner's context.
Foundry Guardrails	Tool call/response interception, PII detection, Task Adherence	Preview	Intercepts tool calls before execution. Blocks PII in outputs. Detects policy drift from system message. Four intervention points : input, output, tool-call (preview), tool-response (preview). Complements MAF validators: Guardrails operate inside the agent executor's GHCP SDK session; MAF validators operate outside, on the typed result.
APIM AI Gateway	Model routing, content safety, token limits	GA	Jurisdiction-based model routing (DE workflow → EU endpoint only). 11m-content-safety policy. Token rate limiting. Semantic caching. All MCP tool calls governed (auth, rate limits, content filtering).
Agent 365 + Entra	Identity, access control, DLP, threat detection	GA May 2026	Per-agent tool access via RBAC on downstream resources. Conditional Access policies on agent identities . Purview DLP on agent interactions . Defender for threat detection .

Policy-as-code: APIM policies version-controlled via APIOps CI/CD. Foundry Guardrails configured per agent. Jurisdiction-specific skills define compliance rules declaratively.

Jurisdiction switching: workflow state carries `jurisdiction` . APIM routes to region-appropriate model endpoint. Foundry Guardrails enforce PII and content safety universally. Jurisdiction-specific skills (right-to-work, works council, GDPR consent) load automatically based on workflow context. Task Adherence guardrail catches drift from jurisdiction policy in the system message.

Audit: [Azure Log Analytics](#) with archive tier (7-12yr retention; immutability enforced via Azure Storage export with immutability policies). Every tool call, model call, enforcement decision, and human interaction logged. Queryable via Log Analytics and [Microsoft Sentinel](#).

10. Builder Experience

Requirement	Approach
Pro-code SDK (Must)	GHCP SDK Python (primary) + TypeScript for skills and MCP servers
Low-code visual builder (Must)	Copilot Studio: visual workflow designer. Supported via Agent 365 (not Foundry Hosted Agents). Custom Control Plane UI natively supports Copilot Studio agents alongside GHCP SDK agents. Not recommended for complex autonomous workflows — available for citizen developers building simpler agent experiences.
Low-code MCP servers	Azure Logic Apps exposed as MCP tools via APIM's REST→MCP gateway . Gives WPP IT teams a no-code path to add new tools (SharePoint → email, Outlook trigger → Dataverse write, prebuilt connectors to 1,400+ systems) without touching Python. Logic Apps is not an orchestrator here — Durable Functions owns workflow state — Logic Apps is purely a tool/integration primitive sitting at the same layer as hand-written MCP servers behind APIM.
Low-code configuration (Must)	Control Plane UI: skill library (backed by Azure API Center), tool catalog (APIM), governance editor, autonomy dials, template fork-and-customise
Agentic builder (Could)	A MAF agent executor generates new SKILL.md files from natural language. Registered in API Center. Human reviews and approves. This capability has been built and demonstrated.
Template library (Should)	Pre-built skill sets and workflow templates in Git. API Center integrates with GitHub Actions for syncing <code>skill.md</code> files from repos. Forkable and customisable.
Knowledge extraction (Could)	Threadlight accelerator: creates MAF workflows with agent executors and skills from interviews and unstructured data. Captures undocumented procedural knowledge from SMEs and produces machine-actionable SKILL.md files.

Skills registered in Azure API Center have lifecycle management (Design → Preview → Production → Deprecated), declared allowed tools, and GitHub Actions integration for `skill.md` files.

11. Control Plane — Two Layers

Foundry Control Plane (platform governance)

Foundry Control Plane provides the platform-level management layer. It is an existing product, not custom build:

- Fleet health dashboards: active agents, run completion rates, compliance posture, cost efficiency, prohibited behaviours

- Agent inventory across all platforms (Foundry-native, custom, external) with lifecycle operations
- Model registry with quota management
- Tool registry with APIM governance
- Continuous evaluation and monitoring with deep links for debugging
- Compliance enforcement (guardrails configuration)
- Integration with Defender (threat detection) and Purview (DLP)

This covers WPP's requirements for agent registry (8.5), model registry (8.1), observability (8.14), agent lifecycle (22.3), and compliance posture monitoring.

Custom Control Plane UI (operator experience)

Foundry Control Plane monitors the platform. WPP's requirements go further — they need an operator experience where 1 human governs 20-50 concurrent workflows with intelligent exception surfacing. This requires a custom React application powered by the Fleet Manager agent:

Capability	Why custom	WPP Refs
Fleet Dashboard (Fleet Manager's assessment)	Foundry shows platform health. WPP needs workflow-level status, SLA tracking, agency/market/jurisdiction filtering. Fleet Manager reasons about what to show.	31.1
Exception-Only Queue	Foundry does not filter to "the 2% needing attention." Fleet Manager composes this intelligently based on business impact, confidence, SLA urgency.	31.2
Instant Situational Awareness	Click any workflow → what agents did, what stopped progress, recommendations, options. Requires OTEL drill-down + Fleet Manager's contextual reasoning. <5 seconds.	31.3
Bulk HITL	Batch similar decisions (e.g., 8 interview schedules), approve in one action. Raises events on all waiting Durable Functions instances.	31.4
Autonomy Dials	Per-workflow threshold sliders (auto-shortlist %, HITL gates). Writes to config store, effective immediately.	21.1
Skill Amplification	Fleet Manager proactively surfaces relevant policy, precedents, recommended approach when operator is uncertain. Powered by Foundry IQ retrieval over WPP corpora.	31.5
Role-Based Operator Views	HR BP sees hiring workflows. Finance BP sees budget gates. IT Ops sees provisioning. RBAC-filtered.	10.1
Cost Dashboard	Per-workflow cost attribution from OTEL + APIM token metrics.	26.4

The custom UI consumes data from: **Application Insights APIs** (agent run traces, token usage, cost, error rates), **Foundry REST APIs** (agent inventory, model deployments), **APIM metrics** (token consumption, rate limit status per model/tool), **workflow state store** (Cosmos DB / Dataverse — phase status, approvals, action ledger), and **Fleet Manager agent assessments** (pushed via SignalR).

12. Voice, Video, Avatar

These are MCP tools and skills invoked from MAF agent executors when a phase needs them.

Capability	Approach
Voice screening	GPT-Realtime as speech-to-speech front end. ACS Call Automation for telephony. MAI-Transcribe-1 for transcription. MAI-Voice-1 for TTS (preview). Tool calls to GHCP SDK backend for reasoning and scoring.
Video meeting notes	Teams bot joins meeting (Bot Framework, GA). Post-meeting transcript via Graph API MCP. A MAF agent executor processes the transcript into structured notes and scores via a downstream validator. Real-time in-meeting speaking achievable via ACS integration but primary value is intelligent note-taking.
Avatar onboarding	An agent executor drafts the personalised script; a downstream deterministic executor invokes the HeyGen API MCP to produce a branded video. Avatars configurable (appearance, voice, branding) and persistent across sessions.

13. Component Summary

Component	Technology	Role
Agent Runtime	GHCP SDK (MIT, Python/TypeScript)	Autonomous reasoning inside MAF agent executors: skills , MCP tools , hooks
Hosting	Foundry Hosted Agents	Containerised deployment, scaling. GHCP SDK wrapped in container exposing Responses API (custom adapter).
Workflow Graph	Microsoft Agent Framework workflows (v1.0 GA, Python/.NET)	Per-phase deterministic graph of typed executors. Plain-function nodes, agent nodes, validator nodes. Pregel BSP execution, fan-out/fan-in, HITL hooks, pause/resume.
Durable Envelope	Azure Durable Functions (GA)	Long-running orchestration: event routing, HITL waits at zero compute (days/weeks), timer escalation, checkpoint/replay, geo-replicated state.
DF ↔ MAF glue	MAF Durable Task extension	Productised pattern wiring MAF workflows as DF activities. Preserves MAF checkpoint state across DF replay.
Fleet Governance	Fleet Manager Agent (GHCP SDK)	Agentic monitoring, exception composition
API & AI Gateway	APIM AI Gateway (GA) + Azure API Center	Unified entry point for models, MCP tools, A2A agents, skills, APIs. Cross-cloud discovery.
Knowledge & Grounding	Foundry IQ + Fabric IQ + Work IQ MCP	Agentic retrieval, business semantic ontology, M365 work graph
Low-code MCP	Azure Logic Apps	Citizen-dev integrations exposed as MCP tools via APIM REST→MCP
Control Plane	Foundry Control Plane (GA)	Agent fleet management, evaluation, compliance
Agent Lifecycle	Agent 365 + Entra Agent ID	Identity, RBAC. GA May 2026 — integration with Hosted Agents needs POC validation. Supports Copilot Studio

		agents natively.
Control Plane UI	Custom React	Fleet dashboard, HITL queue, autonomy dials
Observability	Foundry Tracing (OTEL) + GHCP SDK OTEL	Telemetry, cost attribution
Evaluation	Foundry Evaluators	Task adherence, quality, safety, continuous evaluation
Compliance Layer 1	Foundry Guardrails	Tool call/output interception, PII, task adherence
Compliance Layer 2	APIM AI Gateway	Model routing, content safety, token limits
Compliance Layer 3	Agent 365 + Entra	Identity, access control, DLP, threat detection
Human Surfaces	M365 Agents SDK + Custom UI	Copilot (Teams), Email (Adaptive Cards), Control Plane, Web, Voice
Integration	MCP Servers (via APIM)	Workday, Greenhouse, LinkedIn, ServiceNow, Graph, Dataverse, ACS, HeyGen
Skills	SKILL.md + Azure API Center	Declarative, composable, lifecycle-managed, GitHub Actions integration
Voice	GPT-Realtime + ACS	Speech-to-speech with agentic backend
Audit	Azure Log Analytics	7-12yr retention (immutability via Azure Storage export)
Knowledge Extraction	Threadlight accelerator	Interview-based knowledge capture into SKILL.md files

14. POC 1: Finance Procure-to-Pay

Scope: 30-50 concurrent invoice workflows managed by a Finance Controller via the Control Plane.

Execution shape: Durable Functions orchestration coordinates phase boundaries and HITL waits. Each phase is a MAF workflow graph. Most executors are plain functions; agent executors are used only where reasoning is required.

Phases and executor types:

Phase	MAF executor type	Notes
Intake / OCR	Hybrid	Azure Document Intelligence as a deterministic executor; <code>agent_field_extractor</code> (GHCP SDK) only for low-confidence fields. Validator executor asserts schema before forward.
Three-way match	Deterministic	Plain function: PO / GRN / invoice matching against tolerance rules. No LLM.
GL coding & cost centre	Hybrid	<code>agent_gl_coder</code> reasons over vendor, description, agency context; downstream validator confirms GL exists and cost centre is active.

Routing & approval gate	Deterministic	Threshold-based routing logic. HITL event raised to DF; zero-compute wait.
Payment file generation	Deterministic	Format-driven code. No LLM. Non-revocable action → GHCP SDK hook gates execution until human confirms.
Reconciliation	Hybrid	Deterministic bank statement match; <code>agent_exception_classifier</code> only for unmatched items.

MCP integrations: Workday, Dynamics 365 F&O, Maconomy — all governed through APIM AI Gateway.

Grounding: Fabric IQ for cost-centre / agency hierarchy / budget semantics; Foundry IQ for vendor master data, purchasing policy, and tax-rule corpora.

Demonstrates: deterministic-by-default MAF workflow graph, HITL approval gates (Finance BP interacts via Adaptive Card in Outlook, routed through PA), bulk approval for batched low-risk items, rollback/compensating actions, Fleet Manager monitoring 30-50 concurrent workflows, exception-only Control Plane view, OTEL cost attribution per invoice, Foundry Guardrails inside agent executors, MAF validator executors between agent and non-revocable executors, DF replay across restart.

15. POC 2: HR Talent Lifecycle

Scope: 15-20 concurrent hiring workflows managed by an HR Business Partner via the Control Plane. Five human participants across four timezones.

Execution shape: the 12-week hiring process is a Durable Functions orchestration with ~10 phases. Each phase is a MAF workflow graph — mostly deterministic executors, agent executors where reasoning is genuinely required, validator executors between them. DF owns the long waits between phases; MAF owns the graph shape within a phase.

Phases and executor types:

Phase	MAF executor type	Notes
Budget & approvals	Deterministic	Threshold routing, Fabric IQ lookups. HITL gate to Finance BP via <code>DF_wait_for_external_event</code> .
Job design	Hybrid	<code>agent_jd_drafter</code> (GHCP SDK + Foundry IQ for comp benchmarking); <code>validate_jd_completeness</code> asserts structure.
Sourcing	Deterministic	Greenhouse + LinkedIn MCP queries by criteria. No LLM required.
CV triage / screening	Agentic with validator	<code>agent_cv_scorer</code> reasons over CV; <code>validate_bias_markers</code> downstream executor runs deterministic bias checks and flags to Fleet Manager. Crystallisation target: after N workflows, promote to a deterministic classifier + agent fallback for low-confidence cases.
Voice screening	Agentic	GPT-Realtime + ACS as the executor; structured scoring validator downstream.
Interview coordination	Deterministic	Work IQ for timezone/calendar; plain scheduling logic.

Compliance (jurisdiction-aware)	Hybrid	agent_compliance_narrative for right-to-work / works council interpretation; deterministic rule executors (GDPR consent checklist, EU AI Act classifier); Task Adherence guardrail detects drift from jurisdiction policy.
Offer letter	Hybrid	Template-based deterministic generation; agent_personaliser for narrative sections only. Non-revocable send gated by GHCP SDK hook → human approval.
JML onboarding	Deterministic	ServiceNow ticket creation via MCP. Plain executor.
Avatar welcome video	Hybrid	Agent drafts script; HeyGen MCP deterministic generation.

MCP integrations: Greenhouse ATS, LinkedIn Recruiter, Workday (hiring), Microsoft Graph (calendar/email), ServiceNow (IT provisioning), Azure Communication Services (voice), HeyGen (avatar) — all governed through APIM AI Gateway.

Grounding: Foundry IQ for jurisdiction-specific employment law (US vs DE corpora), GDPR consent guidance, WPP people handbooks; Fabric IQ for headcount, comp bands, agency hierarchy, levelling history (episodic memory of past hires); Work IQ for calendar/timezone/availability and org topology (escalation routing).

Human surfaces: Hiring Manager via M365 Copilot in Teams (agent surfaced via M365 Agents SDK), Finance BP via email Adaptive Cards, candidate via web portal + voice, IT Ops via ServiceNow, HR BP via Control Plane.

Demonstrates: all POC 1 capabilities plus — voice screening with structured scoring, CV parsing with crystallisation pipeline (agent executor → deterministic classifier in API Center), episodic memory from workflow state store + Fabric IQ (recall past hires levelled too low), A2A interop with external candidate agent (governed via APIM), jurisdiction-aware compliance (USA vs Germany enforcement switching via APIM routing + jurisdiction-specific skills + Foundry IQ corpora + Foundry Guardrails), MAF validator executors separating agent judgement from downstream action, autonomy dials (configurable auto-shortlist thresholds), skill amplification (Fleet Manager surfaces policy + precedents via Foundry IQ), process evolution (Fleet Manager proposes crystallisation candidates after completed workflows), synthetic CV evaluation (500 CVs via Foundry Evaluators), avatar onboarding video, **Threadlight** knowledge extraction demo (interview HR SME, produce executable skills).

16. Known Constraints

Constraint	Impact	Mitigation
GHCP SDK in tech preview	API surface may change	Core patterns (skills , MCP , hooks) are GA inside GitHub Copilot's production runtime serving millions of developers daily. SDK is the same code, MIT-licensed.
Microsoft Agent Framework v1.0 released Oct 2025	Framework is young	MAF v1.0 is GA for the core runtime and workflows. Durable task extension for Azure Functions is productised by Microsoft as the Durable Agent Orchestration pattern. Orchestration patterns (sequential, concurrent, handoff, group chat, Magentic-One) are stable. Fallback: the GHCP SDK + Durable Functions combination works without MAF — MAF adds the deterministic graph primitive.
Foundry Hosted Agents: max 5	Scaling ceiling	Multiple deployments, or Azure Container Apps with Foundry telemetry. Preview limit expected to increase at

replicas per deployment (preview)		GA.
Foundry Guardrails tool call interception (preview)	May not be GA for POC	GHCP SDK session hooks provide equivalent enforcement at code level. Guardrails are additive.
APIM A2A agent governance (preview)	A2A features still maturing. A2A is not required for core architecture — only for POC2 external candidate agent demo.	HTTP gateway primitives work today. Purpose-built A2A policies emerging.
Skills in Azure API Center (preview)	Skill registry is new. API Center does not have native Git sync — uses GitHub Actions workflows.	Core skill execution is GHCP SDK native. API Center adds governance layer.
GHCP SDK + Foundry Hosted Agents integration not documented	Hosting adapter needs custom work	Hosted Agents accept any container image that exposes the Responses API protocol. The adapter must translate between Responses API and GHCP SDK session management. This is the primary integration engineering task for the POC.
Agent 365 GA: May 2026	Not yet GA. Integration with Foundry Hosted Agents unclear.	In preview. Whether Hosted Agents auto-register in Agent 365 or require manual onboarding needs validation. Entra Agent ID (the identity layer) is usable independently of Agent 365. Agent 365 natively supports Copilot Studio agents; our Control Plane will also support them.
Foundry IQ / Fabric IQ / Work IQ in public preview	Intelligence layer products are new. APIs evolving.	All three are MCP-addressable — we can fall back to direct Azure AI Search + Fabric SQL + Graph API queries if needed. The IQ products are an upgrade path, not a single point of failure.
MAI-Voice-1	In public preview, not GA. No SLA.	Screening calls use GPT-Realtime (GA) as primary. MAI-Voice-1 is additive for TTS quality.
Copilot Studio on Foundry Hosted Agents	Copilot Studio is not supported on Foundry Hosted Agents.	Copilot Studio agents supported via Agent 365. Custom Control Plane UI natively supports both Copilot Studio and GHCP SDK agents.